



Student perceptions of generative AI tools in higher education: A cross-regional study of use, ethics, and educational utility

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Abstract

Despite the rapid uptake of generative AI (GAI) tools in higher education, there is limited empirical understanding of how students balance perceived educational utility with ethical concerns such as privacy, bias, and overreliance, including risks of cognitive offloading and weakened epistemic authority. This study provides a cross-regional comparison of students' perceptions and use of GAI by focusing on Europe and Australia, two higher education contexts shaped by contrasting policy and institutional cultures, with Europe often characterised by stronger regulatory attention (e.g., data protection and academic integrity governance) and Australia by comparatively pragmatic, implementation-oriented approaches. Using a cross-sectional mixed-methods design, we surveyed 788 university students (Europe $n=402$, Australia $n=386$) via an online questionnaire and analysed the data using ANOVA and chi-square tests, complemented by qualitative content analysis of open-ended responses. Across both regions, students reported strong perceived benefits, most notably accessibility to a fast knowledge base (72.5%) and support for personalised learning (66.4%). Importantly, ethical concerns were widely acknowledged but did not consistently deter adoption, suggesting a pattern of risk normalisation and pragmatic prioritisation of efficiency. However, the results also reveal a statistically meaningful nuance: privacy concerns were more clearly linked to willingness to adopt among Australian students, whereas concerns about bias and dependence showed weaker or non-significant relationships with adoption. Qualitative responses help explain these patterns by indicating that students often recognise ethical risks but accept them as manageable trade-offs when academic productivity gains are salient, while also referencing regional differences in integrity and data-protection expectations. The findings provide actionable implications for higher education institutions and policymakers by supporting the need for structured, multidimensional

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AI literacy initiatives that integrate technical, ethical, and critical competencies, and by highlighting the value of context-specific guidance aligned with distinct regulatory and pedagogical cultures. Limitations include self-reported measures, convenience sampling with disciplinary overrepresentation, and aggregation of diverse European countries into a single regional category, which constrain generalisability and call for longitudinal and more differentiated cross-cultural analyses in future work.

Keywords Generative artificial intelligence (GAI) · Large language models (LLMs) · Student perceptions · Ethical concerns · Higher education

1 Introduction

The rapid integration of artificial intelligence (AI) into higher education has reshaped contemporary learning environments by expanding access to information, supporting personalised learning pathways, and enabling real-time academic assistance (Larsen, 2023; Yang et al., 2025). In recent years, generative AI (GAI) tools, particularly large language models (LLMs) such as ChatGPT, have gained exceptional prominence due to their ability to generate human-like responses, assist with complex academic tasks, and support self-directed learning (Kasneci et al., 2023; Li et al., 2025; Rudolph et al., 2023). As a result, GAI tools are increasingly embedded in students' everyday learning practices across disciplines, raising fundamental questions not only about their pedagogical value but also about their ethical, cognitive, and epistemic implications.

Empirical studies have demonstrated that GAI systems can perform at levels comparable to or exceeding those of students in specific academic domains, including business and medical education (Gilson et al., 2022; Kung et al., 2023; Tajik, 2024; Weeks et al., 2024a). While these findings highlight the disruptive potential of GAI for assessment and academic support, they also intensify concerns regarding academic integrity, overreliance on AI-generated outputs, and the shifting boundaries of epistemic authority in higher education (Cassidy, 2023; Schnabel, 2025; Wu & Shi, 2025). In particular, scholars have warned that students may increasingly defer judgment and problem-solving to AI systems, a phenomenon described as cognitive offloading, which may weaken critical thinking, deep learning, and independent knowledge construction over time (Gerlich, 2025; Lewandowsky, 2025). At the same time, issues related to algorithmic bias, data privacy, and AI “hallucinations” continue to challenge the responsible integration of GAI into educational contexts (Turner et al., 2024; Pérez-Escolar et al., 2025; Wang et al., 2023; Wei et al., 2025).

Despite growing ethical awareness in policy and scholarly discourse, emerging evidence suggests that students' actual use of GAI tools is often driven by pragmatic considerations such as efficiency, accessibility, and academic productivity. This tension between ethical concern and continued adoption has led researchers to question whether ethical risks meaningfully shape student behaviour or whether such risks are increasingly normalised in AI-mediated learning environments. Understanding this ethical–pragmatic gap is particularly important, as it directly affects how institutions design AI governance, assessment practices, and AI literacy initiatives.

Existing research on GAI in higher education has predominantly focused on technical performance, instructional affordances, or educator perspectives, while students' subjective perceptions and decision-making processes remain comparatively underexplored. Moreover, many survey-based studies treat ethical concerns, AI literacy, and adoption as isolated dimensions, offering limited insight into how these factors interact in shaping students' engagement with GAI tools. In particular, little is known about how students reconcile perceived educational utility with ethical awareness, or how this balance may differ across educational systems shaped by distinct regulatory, cultural, and pedagogical traditions.

This study addresses this gap by examining students' perceptions, use patterns, and ethical considerations related to GAI tools in higher education through a cross-regional comparison of Europe and Australia. These two contexts provide a theoretically meaningful contrast. European higher education systems operate within a strongly regulated environment, most notably shaped by the General Data Protection Regulation (GDPR) and institutionalised academic integrity frameworks, which foreground data protection, accountability, and ethical compliance. In contrast, Australian universities have generally adopted a more pragmatic and implementation-oriented approach to educational innovation, emphasising flexibility, employability, and applied learning outcomes (Ibrahim et al., 2023; Australian Government Department of Education, 2023). Rather than relying on broad stereotypes, this comparison allows us to explore how different regulatory and pedagogical cultures mediate students' ethical sensitivity, perceived risks, and willingness to adopt GAI tools in practice.

The central research problem addressed in this study is therefore not whether students use generative AI tools, but how they balance high perceived educational utility with ethical concerns that may or may not translate into behavioural restraint. Drawing on literature on technology adoption, AI literacy, and epistemic authority, this study examines whether ethical concerns function as meaningful deterrents to GAI adoption or whether they are subordinated to pragmatic learning priorities. In doing so, the study also considers the role of AI literacy as a multidimensional construct encompassing technical competence, ethical awareness, and critical evaluation of AI-generated content.

By adopting a mixed-methods approach and foregrounding students' perspectives, this research contributes to the literature in three keyways. First, it provides empirical evidence on the ethical–pragmatic tension underlying students' engagement with GAI tools. Second, it offers a theoretically grounded cross-regional comparison that moves beyond descriptive accounts of adoption to examine how regulatory and educational cultures shape student perceptions. Third, it informs institutional and policy discussions by identifying areas where existing ethical frameworks and AI literacy initiatives may not meaningfully influence student behaviour.

Beyond descriptive cross-regional comparison, this study contributes theoretically by positioning regional educational context as a contextual moderator in students' ethical–pragmatic negotiation of generative AI adoption. Rather than treating adoption as culturally neutral, the study conceptualises regulatory and pedagogical environments as shaping how ethical concerns are interpreted, prioritised, or normalised in practice.

The remainder of this paper is structured as follows. Section 2 reviews the literature on generative AI adoption, ethical concerns, and AI literacy in higher education and develops the conceptual foundation for the study. Section 3 outlines the research design, methodology, and data collection procedures. Section 7 presents the empirical results, followed by a discussion of theoretical and practical implications in Sect. 8. The paper concludes by outlining limitations and directions for future research.

2 Review of literature

This section synthesises prior empirical and conceptual research on students' engagement with generative AI (GAI) tools in higher education. Rather than providing a broad overview of AI in education, the review focuses on three interrelated strands that inform the study's conceptual model: perceived educational utility as a driver of adoption, ethical concerns and their behavioural implications (including risk normalisation, epistemic authority, and cognitive offloading), and AI literacy as a mediating and moderating mechanism shaping responsible and critical use. By integrating these strands, the review moves beyond descriptive accounts of GAI adoption and establishes a theoretically grounded foundation for the proposed research framework.

2.1 Generative AI adoption and perceived educational utility

Research on generative AI (GAI) adoption in higher education has consistently highlighted perceived educational utility as a primary driver of student engagement with these tools. Drawing from broader technology adoption literature, perceived utility reflects learners' assessments of whether a technology enhances task performance, learning efficiency, and academic outcomes. In the context of GAI, this includes support for information retrieval, academic writing, problem-solving, and conceptual clarification (Das et al., 2025). Empirical studies across disciplines report that students perceive GAI tools as valuable academic assistants, particularly for reducing cognitive load associated with routine or time-intensive tasks (Kasneci et al., 2023; Larsem et al., 2023; Lu et al., 2025).

Recent survey-based research indicates that students frequently prioritise pragmatic benefits such as speed, accessibility, and convenience when deciding whether to use GAI tools in their studies (Davis et al., 2025; Rudolph et al., 2023; Wang et al., 2025). These findings align with evidence that GAI systems can produce outputs comparable to high-performing students in structured academic tasks, further reinforcing perceptions of usefulness and efficiency (Gilson et al., 2022; Tajik, 2024; Wang, 2024). As a result, adoption decisions are often framed less as ethical or pedagogical choices and more as instrumental responses to academic demands.

However, while perceived utility is widely acknowledged as a key adoption factor, existing studies tend to treat it as a self-evident or unproblematic construct. Much of the literature emphasises positive learning affordances without sufficiently interrogating how reliance on GAI tools may reshape students' learning strategies, epistemic judgments, or depth of engagement. This gap is particularly salient in light of con-

cerns that frequent reliance on AI-generated outputs may encourage surface learning and reduce opportunities for independent reasoning and knowledge construction.

Moreover, adoption-focused studies often examine utility in isolation, without considering how it interacts with ethical awareness or students' broader understanding of AI systems. This fragmented approach limits explanatory power, as it fails to account for situations in which students acknowledge potential risks but continue to adopt GAI tools due to strong perceived benefits. As a result, the literature provides limited insight into the mechanisms through which perceived utility may override ethical hesitation or critical reflection.

In response to these limitations, recent scholarship has begun to call for more integrated models of GAI adoption that situate perceived utility within a broader framework of ethical considerations and AI literacy (Ng, 2021; Ng, 2024; Choi et al., 2023; Nguyen et al., 2023). From this perspective, perceived educational utility remains a central driver of adoption, but its influence cannot be fully understood without examining how students interpret risks, assess AI-generated knowledge, and negotiate trade-offs between efficiency and responsible use. This study builds on this emerging direction by examining perceived utility not as an isolated determinant, but as one component within a multidimensional model of student engagement with generative AI in higher education.

2.2 Ethical concerns, risk normalisation, and epistemic authority

The rapid diffusion of generative AI (GAI) tools in higher education has intensified scholarly debate around ethical risks, particularly those related to data privacy, algorithmic bias, academic integrity, and overreliance on AI-generated content (Wei et al., 2025b). Normative and policy-oriented literature frequently positions these concerns as critical barriers to responsible adoption, emphasising the need for regulatory safeguards, institutional guidelines, and ethical governance frameworks (UNESCO, 2024; Turner et al., 2024). Within this perspective, ethical awareness is often assumed to translate into cautious or restrained use of GAI tools by students.

However, emerging empirical evidence challenges this assumption. A growing number of studies report a disconnect between students' ethical awareness and their actual usage behaviour, suggesting that ethical concerns do not consistently deter adoption (Pérez-Escolar et al., 2025; Davis et al., 2025). Instead, students appear to acknowledge ethical risks while continuing to engage with GAI tools due to strong perceived benefits. This pattern has been described in prior literature as risk normalisation, whereby potential harm becomes accepted as manageable or unavoidable features of everyday technology use, particularly when efficiency and performance gains are salient.

Risk normalisation is closely linked to broader transformations in students' epistemic practices. As GAI tools increasingly generate plausible, fluent, and contextually relevant outputs, students may shift their trust from traditional sources of knowledge toward AI systems, thereby reconfiguring notions of epistemic authority. Rather than serving solely as supplementary aids, GAI tools can assume an authoritative role in shaping interpretations, arguments, and academic decisions. Scholars have cautioned that such shifts may encourage uncritical acceptance of AI-generated

content, especially when students lack robust strategies for verification and evaluation (Lewandowsky, 2025).

Relatedly, concerns have been raised about cognitive offloading, whereby learners delegate cognitive effort to AI systems, potentially reducing opportunities for deep processing, metacognitive engagement, and independent problem-solving (Gerlich, 2025). While cognitive offloading is not inherently detrimental and may support learning when strategically employed, excessive reliance on GAI tools risks transforming learning into a process of optimisation rather than understanding. Importantly, existing studies rarely examine how students themselves perceive or negotiate these cognitive risks in relation to perceived academic utility.

Despite the prominence of ethical discourse in institutional policies, students' ethical reasoning about GAI use often remains situational and pragmatic. Qualitative research suggests that learners tend to differentiate between abstract ethical principles and concrete academic practices, accepting certain risks as justified trade-offs in high-pressure learning environments. In such contexts, ethical concerns related to bias or dependence may be downplayed, while privacy or assessment-related risks receive greater attention, depending on institutional norms and perceived enforcement (Pérez-Escolar et al., 2025).

These dynamics highlight a critical limitation of existing literature: ethical concerns are frequently treated as static attitudes rather than as context-sensitive judgments shaped by educational culture, assessment regimes, and students' understanding of AI systems (Humr & Canan, 2025). As a result, current research provides limited insight into when and why ethical awareness translates into behavioural restraint, and when it is subordinated to pragmatic learning priorities. Addressing this gap requires moving beyond normative assumptions and examining ethical concerns as part of a broader decision-making process that includes perceived utility, epistemic trust, and learners' capacity to critically engage with AI-generated knowledge.

2.3 AI literacy as a mediating and moderating mechanism

Recent scholarship has increasingly emphasised AI literacy as a critical competence for navigating AI-mediated learning environments. Rather than being limited to technical proficiency, AI literacy is commonly conceptualised as a multidimensional construct encompassing technical understanding of AI systems, ethical awareness of their societal and educational implications, and critical capacity to evaluate the reliability and limitations of AI-generated outputs (Ng, 2024). Within higher education, AI literacy is therefore positioned as a foundational capability for responsible, informed, and reflective engagement with generative AI tools.

Importantly, AI literacy plays a central role in shaping how students interpret both the benefits and risks associated with GAI use. Students with higher levels of AI literacy may be better equipped to recognise the probabilistic nature of AI-generated content, identify potential biases or inaccuracies, and apply verification strategies when integrating such outputs into academic work (Dijkstra et al., 2022). In contrast, limited AI literacy may increase the likelihood of uncritical reliance on AI systems, amplifying risks related to epistemic authority and cognitive offloading.

From an adoption perspective, AI literacy does not function solely as an antecedent or outcome of GAI use, but as a mediating and moderating mechanism that conditions the relationship between perceived utility and ethical concerns. When AI literacy is high, students may be more capable of balancing efficiency gains with ethical and epistemic considerations, thereby engaging with GAI tools in a more reflective and selective manner. Conversely, when AI literacy is limited, strong perceptions of utility may dominate decision-making, while ethical risks are normalised or overlooked.

Despite its growing prominence in policy and pedagogical discourse, empirical research examining AI literacy in relation to students' actual adoption behaviour remains limited. Existing studies often discuss AI literacy normatively or at an institutional level, with fewer investigations exploring how students' self-reported literacy relates to their willingness to adopt GAI tools or their interpretation of ethical risks. Moreover, AI literacy is frequently examined independently of contextual factors, such as regulatory environments or institutional cultures, which may shape how literacy translates into practice.

In response to these limitations, recent frameworks advocate for integrating AI literacy into broader models of technology adoption and ethical engagement, rather than treating it as a standalone skill set (UNESCO, 2024; Davis et al., 2025). Aligning with this perspective, the present study conceptualises AI literacy as a key mechanism through which students negotiate trade-offs between perceived educational utility and ethical concerns, while also accounting for the influence of regional educational context. This integrative approach provides a more comprehensive foundation for understanding students' engagement with generative AI in higher education and directly informs the conceptual model presented in the following subsection.

To ensure conceptual clarity in empirical analysis, AI literacy is analytically distinguished from general familiarity with GAI tools. While familiarity refers primarily to exposure and frequency of use, AI literacy encompasses students' self-reported understanding of system limitations, awareness of ethical implications, and reported practices of verification. Given the survey design, AI literacy is operationalised as perceived competence rather than objectively tested knowledge, which has implications for the interpretation of the findings.

2.4 Conceptual model

Building on prior research on technology adoption, AI literacy, and ethical engagement with artificial intelligence in education, this study is guided by a conceptual model that explains students' adoption and use of generative AI (GAI) tools as the outcome of an interaction between perceived educational utility, ethical concerns, and AI literacy, situated within distinct regional educational contexts.

At the core of the model is perceived educational utility, which captures students' assessments of the practical value of GAI tools for learning-related tasks such as accessing information, supporting academic writing, clarifying complex concepts, and enhancing productivity. Consistent with adoption-oriented perspectives, perceived utility is expected to exert a direct positive influence on students' willingness to use GAI tools in higher education.

In parallel, the model incorporates ethical concerns related to generative AI, including data privacy, algorithmic bias, and overreliance on AI-generated content. While prior normative and policy-oriented literature often assumes that such concerns act as barriers to adoption, emerging empirical findings suggest that ethical awareness does not necessarily translate into behavioural restraint. Accordingly, ethical concerns are modelled as a potential, but not dominant, influence on adoption, allowing for the possibility of weak, indirect, or context-dependent effects.

Crucially, the model integrates AI literacy as a mediating and moderating mechanism linking utility perceptions and ethical awareness to actual adoption behaviour. AI literacy is conceptualised as a multidimensional construct encompassing technical understanding of GAI systems, awareness of ethical and legal implications, and the critical capacity to evaluate and verify AI-generated outputs. Rather than treating ethics and literacy as isolated dimensions, the model assumes that AI literacy shapes how students interpret ethical risks and how they balance those risks against perceived benefits in practice.

Finally, the model explicitly accounts for regional educational context as a contextual moderator influencing the relationships among utility, ethical concerns, AI literacy, and adoption. Europe and Australia are treated not as homogeneous cultural entities, but as higher education environments characterised by different regulatory and pedagogical orientations. European contexts are more strongly shaped by formalised data protection and academic integrity frameworks, while Australian contexts are more commonly associated with pragmatic, application-driven approaches to educational innovation. These contextual differences are expected to condition students' ethical sensitivity and the extent to which ethical concerns influence adoption decisions.

Taken together, the conceptual model positions students' engagement with generative AI not as a purely technical or ethical choice, but as a pragmatic negotiation shaped by perceived utility, mediated by AI literacy, and situated within specific regulatory and educational cultures. This framework provides the theoretical foundation for the study's research questions and hypotheses and guides the interpretation of both quantitative and qualitative findings.

While the model outlines potential mediation and moderation relationships, the present study primarily tests associations between perceived utility, ethical concerns, familiarity, and reported adoption attitudes. The full structural relationships proposed in the model warrant future confirmatory and longitudinal analysis.

Figure 1 illustrates the conceptual model guiding this study. Perceived educational utility is modelled as a direct predictor of students' willingness to adopt generative AI tools in higher education. Ethical concerns (including privacy, bias, and overreliance) are represented as a parallel influence that may exert weak or context-dependent effects on adoption. AI literacy serves as a mechanism that shapes how students interpret ethical risks and translate perceived benefits into actual use. Regional educational context (Europe vs. Australia) is included as a contextual moderator influencing the strength and direction of these relationships, reflecting differences in regulatory and pedagogical cultures. The key constructs underpinning the conceptual model, together with their conceptual and operational definitions, are summarised in

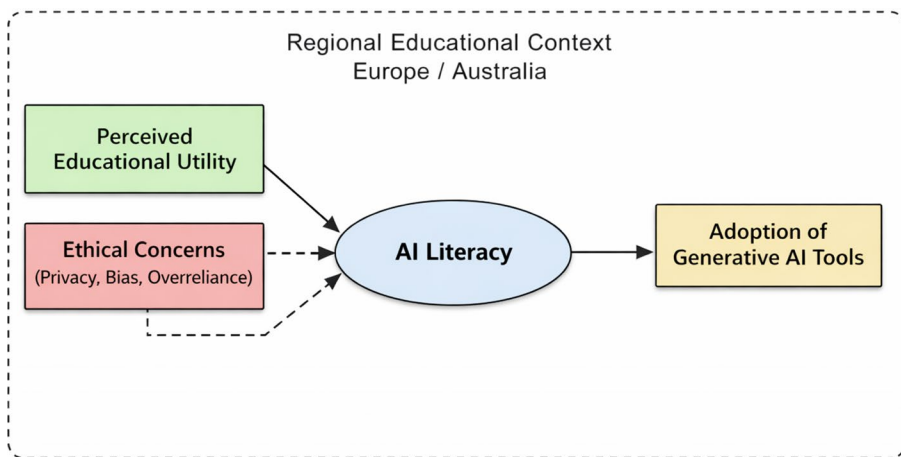


Fig. 1 Conceptual model of students' adoption of generative AI tools in higher education

Table 1. These definitions provide the basis for the operationalisation of variables in the empirical analysis described in the following section.

2.5 Research questions and hypotheses

Building on the conceptual model and the synthesis of prior survey-based research on students' engagement with generative AI (GAI) in higher education, this study adopts a comparative and exploratory research design. The research questions and hypotheses are formulated to examine patterns in students' perceptions, ethical considerations, and readiness to adopt GAI tools across different educational contexts, rather than to establish causal relationships.

2.6 Research questions

RQ1 What patterns characterize students' perceptions of the educational utility and ethical challenges associated with generative AI tools in higher education?

RQ2 How do students' familiarity with generative AI tools and their reported readiness for adoption vary across regional educational contexts (Europe and Australia)?

RQ3 Are there statistically significant differences in perceived educational utility, ethical concerns, and adoption-related attitudes across demographic and regional groups?

2.7 Hypothesis

H1 *Students are expected to report a high perceived educational utility of generative AI tools for learning-related tasks in higher education.*

Table 1 Constructs, conceptual and operational definitions

Construct	Conceptual Definition	Operational Definition (in this study)	Key References	Unit of Analysis/Scope
Perceived Educational Utility	The extent to which students believe that generative AI (GAI) tools enhance their learning effectiveness, efficiency, and academic productivity. This construct reflects pragmatic evaluations of usefulness in academic tasks.	Measured through survey items assessing students' perceived benefits of GAI tools, including access to a fast knowledge base, support for academic writing and research tasks, clarification of complex concepts, and overall learning efficiency. Responses were captured using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree).	Al-Mamary et al. (2024); Davis et al. (2025); Lu et al. (2025); Kasneci et al. (2023)	Individual university students; higher education learning contexts
Ethical Concerns	Students' awareness and evaluation of ethical risks associated with the use of GAI tools, including data privacy, algorithmic bias, academic integrity, and overreliance on AI-generated content.	Operationalised through survey items capturing students' concerns regarding data privacy and security, bias and fairness in AI outputs, and dependence on AI over human judgment in academic work. Items were measured on a 5-point Likert scale.	Turner et al. (2024); Pérez-Escobar et al. (2025); Lewandowsky (2025)	Individual university students; academic use of GAI tools
AI Literacy	A multidimensional competence encompassing students' technical understanding of GAI systems, ethical awareness of their implications, and critical ability to evaluate and verify AI-generated outputs.	Assessed indirectly through items related to students' familiarity with GAI tools, understanding of their limitations, awareness of ethical risks, and reported practices of verification and responsible use. Measured using Likert-type items and qualitative responses.	Ng (2024) ; UNESCO (2024)	Individual university students; digital and ethical literacy in higher education
GAI Adoption/Willingness to Use	The degree to which students are inclined to use or continue using generative AI tools as part of their learning activities in higher education.	Measured through survey items assessing students' willingness to integrate GAI tools (e.g., ChatGPT) into their academic work, perceived appropriateness of use, and intention to rely on such tools for learning-related tasks. Responses were collected on a 5-point Likert scale.	Ibrahim et al. (2023); Rudolph et al. (2023)	Individual university students; voluntary adoption in academic contexts
Regional Educational Context	The regulatory, cultural, and pedagogical environment within which higher education institutions operate, shaping norms, expectations, and governance of AI use in learning.	Treated as a contextual variable distinguishing students studying in European and Australian higher education systems. Used to examine cross-regional differences and moderation effects in the relationships among constructs.	Kasneci et al. (2023); Ibrahim et al. (2023)	Macro-level contextual factor; Europe vs. Australia

H2 *Students are expected to acknowledge ethical concerns related to the use of generative AI tools, while these concerns are not anticipated to uniformly deter their reported willingness to adopt such tools.*

H3 *Higher levels of familiarity with and perceived understanding of generative AI tools are expected to be associated with greater reported readiness to adopt these tools in higher education.*

H4 *There are expected to be statistically significant differences between European and Australian students in ethical concerns and adoption-related attitudes toward generative AI tools.*

H5 *Students are expected to report that generative AI tools are most useful for resolving specific academic issues and supporting project-related work in higher education.*

2.8 Research methodology

This study focuses on the application of GAI tools, particularly LLMs such as ChatGPT, in higher education learning environments. It explores how these tools contribute to personalised learning, problem-solving support, and real-time feedback. Special attention is given to students' perceptions, user experiences, and the ethical and technical challenges related to AI integration in education.

2.9 Scope, aim and research tasks

The primary aim is to conduct a comprehensive analysis of university students' perceptions and experiences regarding the use of GAI tools as learning aids. In addition, the study evaluates their perceived impact on the quality of the learning process and identifies potential barriers to effective and ethical implementation. Based on the findings, the research proposes guidelines for the successful and responsible integration of GAI technologies into higher education. This study aims to:

- Analyse students' perceptions of the benefits and challenges of using GAI tools in academic activities.
- Evaluate the perceived impact of GAI on learning quality and efficiency, particularly in terms of personalised support.
- Identify ethical concerns related to the use of GAI tools, including data privacy, algorithmic bias, and over-reliance on AI-generated outputs.
- Develop recommendations for integrating GAI tools, such as ChatGPT, into existing LMS infrastructures at higher education institutions.
- Compare usage patterns across disciplines and regions, with an emphasis on cross-cultural differences in student attitudes toward GAI.

By addressing these objectives, this study provides insights into how students perceive and utilise GAI tools and similar tools in their educational journeys. The findings aim to inform educators, academic administrators, and policymakers in

designing AI-enhanced learning strategies. Moreover, the study highlights best practices for integrating GAI technologies into pedagogical models while maintaining ethical standards and student well-being.

The integration of GAI into higher education presents both opportunities and challenges. While GAI tools offer students personalised learning support, accessible resources, and interactive problem-solving capabilities, their adoption also raises concerns about academic integrity, data privacy, and potential over-reliance on AI-generated content (Searson et al., 2024). Understanding students' perceptions of these tools is crucial for their responsible implementation in educational settings.

2.10 Research methods used

This study employs a combination of qualitative and quantitative research methods to analyse the role of GAI tools in students' learning experiences and their integration into existing Learning Management Systems (LMS). The selected methods align with the research objectives, focusing on students' attitudes, perceived benefits, ethical concerns, and practical applications of GAI in education.

2.10.1 Core research methods

1. **Theoretical analysis.** A comprehensive literature review was conducted to examine existing research on implementing GAI in higher education. This method provided a foundation for understanding previous findings on student engagement, personalised learning, and ethical considerations such as data privacy and algorithmic bias.
2. **Descriptive Research Method.** A descriptive approach was applied to summarise the data collected through student surveys. This method facilitated the identification of trends in students' attitudes toward GAI tools, enabling a structured comparison of perceived advantages and challenges across different academic disciplines and cultural contexts.
3. **Statistical analysis.** The study employed quantitative data analysis techniques to examine the research questions and hypotheses and identify relationships between key variables. Statistical methods were employed to examine the research questions and hypotheses and to identify patterns and group differences across key variables:
 - **ANOVA (analysis of Variance):** Used to determine significant differences in perceptions of GAI tools across various student groups (e.g., IT vs. social sciences).
 - **Correlation analysis:** Applied to measure the relationship between students' familiarity with GAI tools and their comfort in using them for educational purposes.
 - **Chi-Square Tests:** Used to examine categorical data, particularly related to ethical concerns and adoption rates of AI in education.
 - **Linear Regression:** Implemented to explore associations between independent variables (e.g., prior experience with GAI tools) and students' reported readiness to adopt such tools.

2.10.2 Applied research techniques

- **Survey research**

The primary data collection method involved structured online questionnaires designed to capture students' experiences, attitudes, and behavioural patterns regarding the use of GAI tools. The survey included questions on frequency of use, perceived benefits, ethical concerns, and its role in academic tasks such as research and writing support.

- **Likert scale measurements**

A Likert scale (ranging from 1 to 5) was used to quantify students' attitudes toward the usefulness of GAI tools in education. Respondents evaluated their comfort level, willingness to integrate AI-driven tools into their learning process, and perceived impact on their academic performance.

- **Content analysis**

Qualitative content analysis was conducted on open-ended survey responses to gain deeper insights into students' perspectives on the practical applications and potential risks of GAI tools. This technique enabled a richer understanding of individual concerns and experiences beyond predefined survey categories.

2.11 Description of the survey instrument

To investigate the application of GAI in education, a structured questionnaire was developed to assess students' perceptions, experiences, and attitudes regarding GAI-based tools. The complete survey instrument is provided in Appendix A. It consists of four distinct sections, each addressing a specific aspect of GAI usage in higher education.

1. **Demographic and Academic Background.** Collected data such as students' fields of study, year of study, and prior exposure to AI-related courses.
 2. **Familiarity and Experience with GAI Tools.** Evaluated students' awareness of and engagement with GAI tools.
 3. **Perceptions of GAI in Education.** Questions covered potential benefits, ethical concerns, and recommended strategies for addressing risks.
- **Potential benefits** – personalised learning, enhanced access to academic resources, and real-time assistance (referring to instant academic support, immediate problem-solving, and live feedback on learning tasks).
 - **Ethical concerns** – data privacy, AI bias, and dependency on AI-generated content.
 - **Proposed strategies** – recommendations for mitigating ethical risks and factors

influencing students' willingness to use GAI tools in the future.

4. **Application Potential and Training Needs.** Explored students' readiness for formal training on GAI tools and interest in learning how to integrate these tools into their educational process. It also includes:
 - The perceived impact of GAI on hands-on learning experiences within study programs.
 - Students' willingness to engage in formal training to enhance their understanding and effective use of GAI tools in educational settings.

Before launching the full-scale survey, the questionnaire was pilot tested on a sample of 20 students to evaluate the clarity of the items, the reliability of the instrument, and potential areas for refinement. Reliability analysis was conducted using Cronbach's α coefficient, with all constructs exceeding the 0.7 threshold, indicating a satisfactory level of internal consistency. The pilot study ensured that the final questionnaire provided valid and reliable data for subsequent analysis.

It is important to note that the study captures students' perceived ethical concerns and self-reported adoption intentions rather than observed behavioural constraints. The findings, therefore, reflect subjective evaluations, which may not fully correspond to actual usage patterns in institutional contexts.

2.12 Research sample

The study involved a total of 788 university students, with 402 participants from various European countries and 386 from Australia. Although participants were drawn from multiple European countries, for the purposes of statistical analysis and cross-regional comparison, the data were aggregated into a single European category. This decision was made to ensure a sufficient sample size for regional comparison and to maintain analytical consistency between the two contexts (Europe and Australia).

A convenience sampling method was employed to ensure accessibility while capturing a diverse sample across academic disciplines and educational levels. The sample includes students from undergraduate (first to fourth year), master's (first and second year), and doctoral levels, providing a comprehensive view across different stages of higher education. To facilitate comparative analysis, Europe is treated as a collective entity, acknowledging regional diversity while enabling more robust statistical interpretation. Respondents were further classified by field of study, including Information Technology, Computer Science, Business and Economics, Engineering, Social Sciences, and others. This segmentation enables a multidimensional analysis of students' perceptions, familiarity, and readiness for GAI adoption across both regional and disciplinary contexts.

2.12.1 Classification by year of study

To better understand the academic distribution of participants, students were classified based on their year of study. This study included undergraduate, master's, and doctoral students, with a focus on students from Europe and Australia.

In the European education area, the largest proportion of students are second-year undergraduates, accounting for 39.5% of the sample. The proportions of first-, third-, and fourth-year undergraduate students are relatively consistent, ranging from 15.3% to 16%. Among master's students, 4.3% are in their first year, while 6.3% are in their second year. Doctoral students represent the smallest group, making up just 3.3% of the respondents.

In Australia, the classification included undergraduate students (first to fourth year) and master's students (first and second year). The majority of respondents were master's students, with 31.6% in their first year and 21.2% in their second year. Among undergraduates, the proportions of first- and fourth-year students are similar, at approximately 14%, while second-year students account for 10.6%, and third-year students make up 8%.

2.12.2 Classification by field of study

The participants represent a diverse range of academic disciplines. Among European students, the most dominant fields are Information Technology (IT) and Computer Science, with IT students comprising 38.8% of the sample and Computer Science students accounting for 25.9%. Students from Business and Economics make up 13.7% of respondents, while Engineering students represent 11.4%. Architecture students account for 6% of the sample, and Social Sciences students are the least represented, at 1.7%. Additionally, 2.5% of surveyed students do not belong to any of the specified fields.

The distribution of Australian students by field of study incorporated additional categories such as Medicine & Health and Accounting. In the observed sample of Australian students, as in the European sample, respondents from the Information Technology study program are the most dominant, accounting for 54.5% of the total. Business and Economics students represent the second-largest group, making up 22.9% of the sample. The share of students from other fields is significantly smaller, ranging from 1.1% to 7.4%, with Engineering students being the most represented among them at 7.4%.

These classifications offer valuable insights into the demographic and academic diversity of the research sample, facilitating a comparative analysis of GAI perceptions across various disciplines and educational systems.

2.13 Research sample

This section presents the findings obtained from a survey conducted among students at various European universities. The study encompassed students from diverse academic disciplines, including Information Technology, Computer Science, Social Sciences, and Business Economics, providing insights into different levels of famil-

ilarity and perspectives on GAI technologies. The results are analysed concerning the formulated hypotheses, with statistical tests applied to validate or refute initial assumptions.

2.13.1 Results for students from Europe

A detailed analysis of the survey responses from 402 European students is presented below, with a comparison between the proposed hypotheses and observed results.

1. Familiarity with GAI and field of study

- **H₁**: Students in Information Technology and Computer Sciences are more likely to be familiar with GAI systems compared to students in Social Sciences or Business Economics.

To examine differences in familiarity with GAI tools across academic disciplines, an **ANOVA test** was conducted. Familiarity with GAI systems was defined as the dependent variable and categorised into four levels: '*Not at all familiar*', '*Somewhat familiar*', '*Very familiar*', and '*Expert*'.

The **independent variable** was the **field of study**, which consisted of nine categories. Table 2 presents the results of the ANOVA test conducted to assess whether there is a statistically significant difference among groups.

Since $p=0.000 < 0.05$, the ANOVA test confirms a statistically significant difference in familiarity levels with GAI systems across different academic fields. Consequently, a post-hoc analysis was conducted to determine which specific groups differed significantly.

Post-hoc test results The post hoc comparison (Table 3) was conducted using **Tukey's HSD test** to identify differences between groups.

Based on the analysis of the results presented in Table 3, a statistically significant difference in familiarity with GAI systems was observed among students of Information Technology, Business and Economics, Engineering, and Architecture. Regarding computer science students, a statistically significant difference was found between students in engineering and architecture.

The effect size for this analysis was calculated using η^2 :

Table 2 ANOVA test analysis of differences in knowledge of the GAI system among students

ANOVA					
<i>How familiar are you with GAI systems, like ChatGPT, and their applications in learning?</i>					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	23.198	6	3.866	9.471	0.000
Within Groups	161.240	395	0.408		
Total	184.438	401			

Table 3 Post-hoc test to examine differences in knowledge of the GAI system among different fields of study

(I) Please specify your learning area:	(J) Please specify your learning area:	Mean Difference (I-J)	Std. Error	Sig.
Information technology	Computer Science	0.141	0.081	0.587
	Business and Economics	0.408*	0.100	0.001
	Engineering	0.503*	0.107	0.000
	Architecture	0.782*	0.140	0.000
	Social Sciences	-0.016	0.247	1.000
	Other	0.599	0.208	0.064
Computer Science	Information technology	-0.141	0.081	0.587
	Business and Economics	0.267	0.107	0.160
	Engineering	0.362*	0.113	0.025
	Architecture	0.641*	0.145	0.000
	Social Sciences	-0.157	0.249	0.996
	Other	0.458	0.212	0.318
Business and Economics	Information technology	-0.408*	0.100	0.001
	Computer Science	-0.267	0.107	0.160
	Engineering	0.095	0.128	0.990
	Architecture	0.374	0.156	0.204
	Social Sciences	-0.423	0.256	0.649
	Other	0.191	0.220	0.977
Engineering	Information technology	-0.503*	0.107	0.000
	Computer Science	-0.362*	0.113	0.025
	Business and Economics	-0.095	0.128	0.990
	Architecture	0.279	0.161	0.593
	Social Sciences	-0.519	0.259	0.416
	Other	0.096	0.223	1.000
Architecture	Information technology	-0.782*	0.140	0.000
	Computer Science	-0.641*	0.145	0.000
	Business and Economics	-0.374	0.156	0.204
	Engineering	-0.279	0.161	0.593
	Social Sciences	-0.798	0.274	0.059
	Other	-0.183	0.240	0.988
Social Sciences	Information technology	0.016	0.247	1.000
	Computer Science	0.157	0.249	0.996
	Business and Economics	0.423	0.256	0.649
	Engineering	0.519	0.259	0.416
	Architecture	0.798	0.274	0.059
	Other	0.614	0.315	0.448
Other	Information technology	-0.599	0.208	0.064
	Computer Science	-0.458	0.212	0.318
	Business and Economics	-0.191	0.220	0.977
	Engineering	-0.096	0.223	1.000
	Architecture	0.183	0.240	0.988
	Social Sciences	-0.614	0.315	0.448

$$\eta^2 = \frac{23.198}{184.438} = 0.13$$

The results indicate that students in Information Technology demonstrated the highest familiarity with GAI systems, significantly surpassing students in Business & Economics, Engineering, and Architecture. Similarly, Computer Science students exhibited greater familiarity than Engineering and Architecture students, but their familiarity levels were not significantly higher than those of Business & Economics or Social Sciences students.

Conversely, Architecture students recorded the lowest familiarity levels, with a statistically significant difference compared to students in IT, Computer Science, and Business & Economics.

Figure 2 presents the average familiarity levels with GAI systems across different fields of study. The highest levels of familiarity were observed among Information Technology and Social Sciences students, while Business & Economics and Engineering students occupied a mid-range position. The relatively high familiarity among social science students may reflect curricular innovations that increasingly incorporate data analytics, digital communication, and research methods requiring AI literacy (Zeng et al., 2024). Architecture students demonstrated the lowest familiarity, reinforcing the statistical significance found in the ANOVA and post-hoc analysis.

The η^2 value of 0.13 suggests a moderate to high effect size, confirming that the field of study has a significant influence on students' familiarity with GAI tools. The findings related to H1 indicate that students in Information Technology demonstrate significantly higher familiarity with GAI systems compared to students in Business & Economics, while differences involving other fields are less consistent. The findings related to H1 suggest that the proposed relationship holds for certain student subgroups, whereas it is not uniformly observed across all comparisons.

2. User's knowledge of GAI applications and comfort in using them

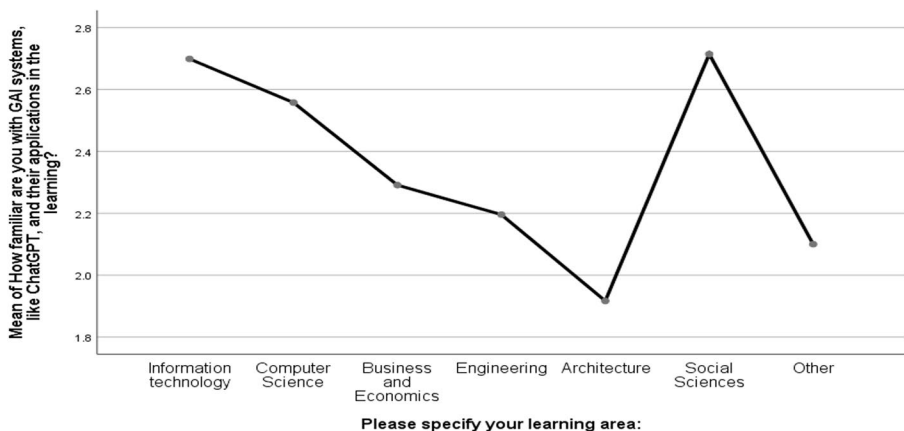


Fig. 2 Average values of knowledge of the GAI system by field of study

Table 4 Contrast coefficients for different fields of study in the analysis of differences in familiarity with GAI

Contrast Coefficients							
Contrast	Please specify your learning area:						
	Information technology	Computer Science	Business and Economics	Engineering	Architecture	Social Sciences	Other
1	1	1	-1	-1	0	1	-1

Table 5 Contrast test results for different fields of study

	Contrast	Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
Assume equal variances	1	-2.08	0.542	-3.836	395	0.000
Does not assume equal variances	1	-2.08	0.390	-5.341	25.559	0.000

• **H₂**: Individuals who are more familiar with GAI systems will show a higher level of comfort in using such technology for educational purposes.

After testing H1, which confirmed that students in Information Technology and Social Sciences exhibit the highest familiarity with GAI systems, we now analyse their level of comfort in integrating such technology for educational purposes. Additionally, Computer Science students are included in the comparison, as they ranked third in familiarity.

To evaluate whether familiarity with GAI systems translates into greater comfort in using these technologies, we employed contrast coefficients in an ANOVA test. This method allows us to compare multiple groups simultaneously and determine whether differences in comfort levels are statistically significant. Since Table 2 demonstrated the presence of seven groups (Information Technology, Computer Science, Business & Economics, etc.), the contrast coefficient approach is statistically valid for this analysis.

Table 3 presents the contrast coefficients, which were used to assess differences in comfort levels among students from different academic backgrounds.

The contrast coefficients in Table 4 indicate a significant difference in the level of comfort in using GAI systems between students more familiar with such technologies (Information Technology, Computer Science, and Social Sciences) and those less familiar (Business & Economics, Engineering, and Other Fields). Architecture students were excluded from this classification and treated as neutral. The contrast test results in Table 5 confirm whether these differences are statistically significant.

Since $p < 0.05$, we confirm the presence of **statistically significant differences** in comfort levels between student groups. Students in Information Technology and Computer Science demonstrated significantly higher comfort levels in using GAI tools compared to Business & Economics students. Social Sciences students, despite their high familiarity with GAI, did not exhibit a statistically significant difference in comfort levels compared to other groups. Architecture students were classified as neutral, indicating that their comfort levels neither significantly align with highly familiar nor less familiar groups.

The analysis confirms that students from technical fields (Information Technology and Computer Science) exhibit a higher level of comfort in integrating GAI tools into their educational activities. However, the assumption that Social Sciences students would also show significantly higher comfort levels was not supported. The findings related to H2 suggest that higher familiarity with GAI tools is associated with greater comfort among students in technical disciplines, while this pattern is less evident among students in the Social Sciences.

5. Perception of the benefits of use and ethical interests

- **H₃:** Respondents will highlight convenience, personalised learning, and accessibility as key benefits of using GAI tools in education.

To examine this hypothesis, we conducted a frequency analysis to determine how many students identified convenience, personalised learning, and accessibility as key benefits of GAI tools in education. This method allowed us to observe the prevalence of each benefit among respondents. Graphs are provided for better visualisation of frequency distributions.

The frequency analysis results for each benefit are summarised in Supplementary Tables B1 – B7 (Appendix B).

- **Accessibility** was the most frequently mentioned advantage, with 70.9% (285 respondents) selecting “*Access to a fast knowledge base*” as a significant benefit (Table B2).
- **Personalised learning** was identified as important by 44.8% (180 respondents) (Table B3).
- **Convenience**, measured as “*Real-time feedback and assessment*”, was less frequently acknowledged, with only 31.8% (128 respondents) considering it a major advantage (Table B1).

Other perceived advantages showed lower levels of support:

- “Improved engagement and motivation” – 21.6% (Table B4)
- “Improved learning outcomes” – 27.4% (Table B5)
- “Improved productivity and application of technology” – 48.0% (Table B6)

To determine whether the observed differences are statistically significant and to assess the validity of H3, we conducted a binomial test. This test determines whether the proportion of respondents identifying specific advantages significantly deviates from the expected 50% threshold. Given that H3 states that respondents will emphasise convenience, personalised learning, and accessibility as key advantages, we classified affirmative responses (“yes”) as “success” and negative responses (“no”) as “failure.” In SPSS, data were coded as 0 (no) and 1 (yes) for statistical analysis. The binomial test was conducted with a test proportion of 0.50 (50%), representing the assumption that respondents would be equally likely to select or reject a given advantage. The test results are summarised in Table B7.

1. **Convenience (Real-time feedback and assessment) was NOT confirmed as a key advantage**

- Only 31.8% of respondents considered it a major benefit.
- Binomial test: $p < 0.000$, significantly below the 50% threshold → Rejected as a key advantage.

2. **Personalised learning was NOT strongly confirmed**

- Identified by 44.8% of respondents, slightly below the 50% threshold.
- Binomial test: $p = 0.041$ → Weak evidence for personalised learning as a primary benefit.

3. **Accessibility was STRONGLY confirmed as a key advantage**

- 70.9% of respondents highlighted “*Access to a fast knowledge base*”.
- Binomial test: $p < 0.000$ → Statistically significant and valid.

Based on these findings, accessibility is confirmed as a key benefit, while convenience and personalised learning are not strongly supported. These results suggest that students primarily value GAI tools for providing quick access to knowledge rather than for improving engagement or real-time feedback. Further qualitative research may be necessary to explore why personalised learning was not as strongly validated despite its perceived importance in AI-enhanced education.

1. Perception of the benefits of use and ethical interests

- **H₄**: Ethical concerns such as data privacy, bias, and overreliance on artificial intelligence will be major concerns regarding introducing the GAI tool into education.

To evaluate this hypothesis, we conducted a frequency analysis to assess students’ perceptions of key ethical concerns:

1. Bias and fairness in decision-making
2. Privacy and data security concerns
3. Dependence on technology over human interaction

The frequency analysis results provide insights into how often each ethical concern was raised among respondents. These findings help determine whether concerns about the ethical implications of GAI tools, significantly impact students’ willingness to adopt such technologies in educational contexts.

The frequency analysis results for each ethical concern are summarised in Supplementary Tables B8 – B10 (Appendix B).

- **Bias and fairness in decision-making** was the least frequently cited concern,

with only 29.4% (118 respondents) identifying it as an issue, while 70.6% (283 respondents) did not perceive it as a major problem (Table B8).

- **Privacy and data security concerns** were more evenly split, with 48.6% (195 respondents) expressing concern and 51.4% (206 respondents) not seeing it as a significant issue (Table B9).
- **Dependence on technology over human interaction** was also evenly divided, with 49.6% (199 respondents) perceiving it as a concern, while 50.4% (202 respondents) did not (Table B10).

These results suggest that while bias and fairness are not a predominant concern, privacy and overreliance on technology are more polarizing issues, indicating a divided perspective among students regarding these ethical considerations.

To further assess the significance of these concerns, we conducted a Chi-Square test, which evaluates whether there is a statistically significant association between ethical concerns and students' willingness to use GAI tools in education. The Chi-Square results are summarised in Supplementary Tables B11 – B13 (Appendix B).

Chi-Square Test for Privacy and Data Security Concerns (Table B11)

- $p=0.592$ No statistically significant association between privacy concerns and willingness to use GAI tools.

Chi-Square Test for Bias and Fairness in Decision-Making (Table B12)

- $p=0.390$ No statistically significant association between concerns about bias and fairness and willingness to use GAI tools.

Chi-Square Test for Dependence on Technology Over Human Interaction (Table B13)

- $p=0.970$ No statistically significant association between concerns about technology dependence and willingness to use GAI tools.

The relatively low concern about algorithmic bias, identified by only 29.4% of respondents, suggests that students may either underestimate the risks associated with AI-driven decisions or prioritise immediate educational benefits over ethical considerations.

1. **Bias and fairness in decision-making were not a major concern for most students.**
 - Only 29.4% identified bias as a concern, while 70.6% did not consider it a significant issue (Table B12).
 - The Chi-Square analysis ($p=0.390$) confirmed no significant relationship between concerns about bias and students' willingness to use GAI tools.

2. Privacy and data security concerns were more divided, but still not statistically significant.

- 48.6% expressed privacy concerns, while 51.4% did not (Table B13).
- The Chi-Square test ($p=0.592$) showed no statistically significant association between privacy concerns and the adoption of GAI tools.

3. Dependence on technology was also divided but showed no statistical impact.

- 49.6% of students expressed concern about overreliance on AI, while 50.4% reported no such concern (Table B14).
- The Chi-Square test ($p=0.970$) indicated no statistically significant relationship between concerns about AI dependence and students' willingness to use GAI tools in educational contexts.

The statistical analyses did not support a significant association between ethical concerns and students' willingness to adopt GAI in education. While privacy and AI dependency concerns were more evenly distributed among respondents, they did not have a measurable statistical impact on students' perceptions. Concerns related to bias and fairness emerged as the least prominent, with the majority of students not perceiving these issues as major barriers. These results suggest that although ethical concerns regarding GAI tools exist, they are not dominant factors influencing students' willingness to integrate tools like ChatGPT into their learning processes. Further research may be warranted to explore contextual or regional variations in the perception of ethical risks, as students from different disciplines or cultural backgrounds may evaluate these concerns differently.

Although chi-square analysis revealed no statistically significant association between ethical concern levels and actual adoption of GAI tools, qualitative responses provide valuable nuance to this finding. Many students expressed an awareness of ethical risks but emphasized pragmatic or utilitarian motives for continued use.

For instance, one European student remarked, *"I know AI can be biased, but it saves me hours of work, and I always double-check what it gives me."* Similarly, an Australian participant noted, *"Privacy is a concern, but for learning purposes, it's worth the trade-off."* These statements suggest that while ethical considerations are recognised, they are often subordinated to perceived efficiency, accessibility, and academic utility.

1. Perception of the benefits of use and ethical interests

- **H₅:** Participants will find GAI tools most useful in providing immediate help and assistance for specific questions within certain subjects and in supporting research tasks within the educational program.

To evaluate this hypothesis, we conducted a frequency analysis to assess students' perceptions of GAI tools' usefulness in education. The analysis focused on responses to the following questions:

1. *“In what areas of an educational program do you think GAI tools could be most useful?”*
2. *“In what ways do you believe GAI tools could be most beneficial for improving experiential learning?”*

The results provide insights into how students perceive GAI tools' role in academic tasks, research support, and experiential learning.

The frequency analysis results for each question are summarised in Supplementary Tables B14 - B16 (Appendix B).

1. Areas Where GAI tools are Most Useful (Table B14)

The most frequently cited applications of GAI tools were:

- Writing assignments (e.g., term papers) – 19.7% (77 respondents)
- Preparing projects – 13.0% (51 respondents)
- Experiential learning – 7.4% (29 respondents)

Additional responses, such as team collaboration (5.6%) and mixed academic support tasks highlight students' recognition of GAI as a practical aid in structured academic contexts.

2. Ways GAI Tools Supports Experiential Learning (Table B15)

The most frequently mentioned uses of GAI tools in experiential learning were:

- Providing examples – 14.4% (57 respondents)
- Generating questions – 8.1% (32 respondents)
- Creating scenarios – 6.3% (25 respondents)
- Customising learning materials – 8.9% (35 respondents)

These findings suggest that students perceive GAI tools, such as ChatGPT, as valuable for enhancing learning through examples, interactive question generation, and content adaptation rather than solely as an assignment-writing tool.

To further validate the results, a Chi-Square test was conducted to assess whether there was a significant relationship between the perceived usefulness of GAI tools in different educational contexts and its ability to improve experiential learning. The results are presented in Table B16.

The Pearson Chi-Square value of 852.449 ($p < 0.001$) indicates a statistically significant relationship between the perceived usefulness of GAI tools and their effectiveness in enhancing experiential learning. This confirms that students believe such tools can play a meaningful role in providing immediate assistance for specific subjects and supporting research tasks within the educational program.

1. **Students primarily perceive GAI tools as useful for academic writing and research-related tasks.** Writing assignments (19.7%) and project preparation

(13.0%) were the most frequently cited applications, suggesting that students recognize these tools as valuable for structured academic work.

2. **Students view GAI as beneficial for interactive and experiential learning.** Providing examples (14.4%) and generating questions (8.1%) were among the most frequently mentioned features, indicating that GAI tools support engagement in interactive learning processes.

The statistical results confirm Hypothesis 5. The Chi-Square analysis ($p < 0.001$) supports the existence of a significant relationship between perceived usefulness and the potential of GAI tools to improve experiential learning. These findings suggest that students value GAI not only for content generation (e.g., writing assignments), but also for supporting critical thinking and active learning (e.g., generating questions, providing examples, Customising materials).

Students identify GAI tools as particularly useful for providing timely academic support and enhancing research-related activities. The results highlight the importance of integrating these tools responsibly into educational programs, ensuring their use as learning aids rather than as mechanisms for bypassing critical engagement.

Future research could explore disciplinary differences in the perceived usefulness of GAI tools and examine the long-term effects of integrating AI technologies into experiential learning models.

2.13.2 Results for students from Australia

Headings of sub-sections should be in Times New Roman 10-point bold in sentence structure (Sub-Section Heading Style). This section presents the findings from a survey conducted among 386 students across various Australian universities. The research mirrors previous studies conducted among European students, facilitating comparative analysis to discern similarities and differences in student perspectives across continents.

1. Familiarity and study area

- **H₁:** Students in Information Technology and Computer Sciences are more likely to be familiar with GAI systems compared to students in Social Sciences or Business Economics.

To assess this hypothesis, we analysed the level of familiarity with GAI tools among students from various disciplines. Familiarity was categorised into four levels: “Not at all familiar,” “Somewhat familiar,” “Very familiar,” and “Expert.” The field of study served as the independent variable. An analysis of Variance (ANOVA) test was conducted to determine if there were statistically significant differences in familiarity levels across different study areas. The detailed ANOVA results are presented in Supplementary Table B17 (Appendix B).

The ANOVA results indicate a p-value of 0.195, suggesting no statistically significant differences in familiarity with GAI systems among students from different fields of study. To further explore this finding, a Chi-Square test was performed to examine

the relationship between the field of study and the level of familiarity with GAI technologies. The corresponding Chi-square test results are summarised in Supplementary Table B18 (Appendix B).

The Chi-Square test results show a Pearson Chi-Square value of 27.455 with 27 degrees of freedom and a p-value of 0.439. Since the p-value is greater than 0.05, we conclude that there is no statistically significant relationship between the field of study and the level of familiarity with GAI systems. Therefore, hypothesis H1 is not supported; students in Information Technology and Computer Sciences are not significantly more familiar with GAI systems like ChatGPT compared to students in Social Sciences or Business Economics.

These findings align with recent studies indicating that familiarity with GAI tools is becoming widespread across various disciplines. For instance, a national survey of Australian university students revealed that students across diverse fields are engaging with GAI technologies, reflecting a broadening familiarity beyond traditional tech-focused disciplines.

2. User's knowledge of GAI applications and comfort in using them:

- **H₂:** Individuals who are more familiar with GAI systems will show a higher level of comfort in using such technology for educational purposes.

In this hypothesis test, we aimed to determine whether students exhibit varying levels of comfort in using GAI technologies for educational purposes. To analyse this, we employed contrast coefficients within an ANOVA framework, which is appropriate when comparing more than two groups. As shown in Supplementary Table B19 (Appendix B), our study encompassed seven distinct fields of study, Information Technology, Computer Science, Business and Economics, Engineering, Social Sciences, and others, thereby satisfying the requirement for multiple groups in the ANOVA test.

Contrast coefficients are instrumental in comparing specific group combinations within an ANOVA test. By assigning values such as 1 and -1 to these coefficients, we can determine which particular group comparisons to perform, as presented in Supplementary Table B19 (Appendix B). This method allows for targeted analysis of differences between selected groups.

Based on the results presented in Supplementary Table B20 (Appendix B), the p-values are 0.574 under the assumption of equal variances and 0.555 when equal variances are not assumed. Since both p-values exceed the conventional threshold of 0.05, we conclude that there is no statistically significant difference in comfort levels among groups defined by their familiarity with GAI systems. The contrast value of 0.28 indicates a minor difference between groups; however, this difference lacks statistical significance. Consequently, there is insufficient evidence to assert that students more familiar with GAI systems exhibit higher comfort levels when using this technology for educational purposes than less familiar students.

3. Perception of benefits of use and ethical concerns:

- **H₃**: Respondents will highlight convenience, personalised learning, and accessibility as key benefits of using GAI tools in education.

To examine hypothesis H₃, we conducted a frequency analysis, similar to the approach used with European students. Frequency analysis involves counting the occurrences of each value or category within our data, making it a suitable method for our dataset.

Supplementary Table B22 (Appendix B) presents the analysis for “Personalised learning experience,” where 42.5% of respondents acknowledged it as a benefit. While this result highlights some recognition of personalised learning as an advantage, the percentage remains below the expected 50% threshold.

Supplementary Table B23 (Appendix B) displays the results for “Real-time feedback and assessment.” Only 36.3% of respondents identified this feature as a key benefit, suggesting that convenience was not widely recognized as a primary advantage.

Other perceived benefits received limited support:

- **Improved engagement and motivation**: 26.9% of respondents identified this as a benefit Supplementary Table B24 (Appendix B).
- **Improved learning outcomes**: Recognized by 42.7% of respondents with mixed perceptions Supplementary Table B25 (Appendix B).
- **Improved productivity and application of technology**: Identified by 35% of respondents as a valuable feature (Supplementary Table B26 (Appendix B)).

To perform a binomial test after the frequency analysis, we need to have data representing two categories, such as “yes” and “no” (In our case, in the SPSS program, the data are labelled with 0 and 1, which can also be seen in the previous frequency analyses). This means that we can perform a binomial test. The binomial test checks whether the frequency of a category is significantly different from the expected one, i.e. whether the frequency is greater than 50%, and we need this to see if any of the categories stand out. To accurately assess the validity of the hypothesis, a binomial analysis was also conducted, as shown in Supplementary Table B27 (Appendix B).

To assess whether these observed proportions are statistically significant, a binomial test was conducted. This test determines whether the percentage of respondents identifying specific advantages deviates significantly from the expected 50% threshold.

- **“Real-time feedback and assessment”**: 36.3% ($p < 0.000$) → Significantly below the expected threshold, confirming that convenience was NOT identified as a dominant benefit.
- **“Personalised learning experience”**: 42.5% ($p = 0.004$) → Slightly lower than the expected 50%, indicating weak evidence supporting personalised learning as a primary advantage.
- **“Access to a fast knowledge base”**: 46.6% ($p = 0.203$) → While notable, this result is not significantly higher than the expected threshold, suggesting accessibility is perceived positively but not overwhelmingly.

While accessibility and personalised learning were positively perceived, neither achieved sufficient statistical confirmation to be considered dominant advantages. These findings highlight that while students see some benefits in using GAI tools, these features are not as strongly emphasised as anticipated.

4. Perceived benefits of use and ethical concerns:

• **H₄:** Ethical concerns such as data privacy, bias, and overreliance on AI will be the main concerns regarding the introduction of GAI tools in education.

To examine this hypothesis, which explores whether ethical concerns like data privacy, bias, and overreliance on AI are the primary concerns regarding GAI systems integration in education, a frequency analysis was conducted. This method is suitable for identifying how often each category appears in the dataset.

Supplementary Table B28 (Appendix B). summarises the frequency analysis for “Bias and fairness in decision-making.” The results indicate that 69.4% of respondents did not consider bias and fairness to be a significant concern, while 30.6% expressed concern on this issue.

Supplementary Table B29 (Appendix B). presents the results for “Dependence on Technology over Human Interaction,” where 55.4% of respondents identified this as a significant concern, while 44.6% did not regard it as an issue.

Supplementary Table B30 (Appendix B). details the analysis for “Privacy and data security concerns.” Here, opinions were almost evenly split, with 49.5% of respondents expressing concern and 50.5% not viewing this issue as a primary concern.

The frequency analysis results indicate that dependence on technology and data privacy are notable concerns for a considerable portion of respondents. In contrast, concerns about bias in decision-making were less prevalent.

Chi-square tests were conducted to determine whether these concerns were significantly associated with respondents’ willingness to use GAI tools:

- **Bias and fairness in decision-making:** The Chi-square test did not reveal a statistically significant association ($p=0.439$) (Supplementary Table B31 (Appendix B)).
- **Dependence on technology over human interaction:** The Chi-square test also showed no significant association with GAI adoption ($p=0.539$) (Supplementary Table B32 (Appendix B)).
- **Privacy and data security concerns:** The Chi-square test identified a statistically significant relationship between privacy concerns and willingness to use GAI tools ($p=0.004$) (Supplementary Table B33 (Appendix B)), indicating a meaningful connection between these variables.

While concerns about bias and technology dependence did not significantly influence students’ willingness to use GAI tools, privacy and data security concerns emerged as important factors that impacted their willingness to adopt this technology. This highlights the need to address privacy-related concerns to encourage broader acceptance of GAI tools in educational settings.

5. Potential Applications, Usefulness in Educational Programs, and Effectiveness:

- **H₅:** Participants will find GAI tools most helpful in providing immediate help and assistance for specific questions within certain subjects, as well as in supporting research tasks within the educational program.

To examine this hypothesis, a frequency analysis of responses to survey questions was conducted, following the same procedure applied in the analysis of responses from European students. The analysis focused on identifying the areas in which GAI tools are considered most useful, as well as their perceived potential to enhance experiential learning.

Supplementary Table B34 (Appendix B) illustrates the frequency analysis for the question: “In what areas of an Educational Program do you think GAI tools could be most useful?”

- **Experiential learning** was the most frequently cited benefit, with 20.5% of respondents highlighting this category.
- **Writing assignments** emerged as another notable area, with 11.4% identifying this as a key benefit.
- **Preparing projects** was also considered significant, cited by 8.3% of respondents.

These findings suggest that GAI tools are primarily valued for their ability to assist with practical engagement and creative tasks within educational programs.

Supplementary Table B35 (Appendix B) outlines responses regarding role of GAI tools in improving experiential learning:

- **Providing examples** was identified by 18.1% of respondents as the most effective use of GAI tools for enhancing experiential learning.
- **Generating questions, creating scenarios, and Customising materials** were mentioned by 10.1% of participants as valuable features.

This result indicates a statistically significant association between participants’ perceptions of the usefulness of GAI tools in educational tasks and their perceived effectiveness in enhancing experiential learning.

To further evaluate the hypothesis, a Chi-square test was conducted to assess the statistical significance of the observed trends.

Table 40 presents the Chi-square test results, where:

- The Pearson Chi-square value was 622.432, with 256 degrees of freedom (df) and a p-value of <0.001.
- This result indicates a statistically significant association between participants’ perceptions of GAI usefulness in educational tasks and its ability to enhance experiential learning.

Based on the frequency and Chi-square test results, **H₅ is confirmed**. Respondents believe that GAI tools are particularly valuable for providing immediate assistance

Table 6 Comparative analysis of the results obtained for Europe and Australia

Hypothesis	Europe - results	Australia - results
H1: Students in IT and computer science majors will be more familiar with GAI systems than others.	No statistically significant difference was found between computer science students and other fields.	No statistically significant difference was found between computer science students and other fields.
H2: Greater familiarity leads to greater comfort in using GAI systems in education.	IT students demonstrated greater comfort.	No significant difference in comfort levels between groups was found.
H3: Students will highlight convenience and accessibility as key advantages of GAI tools.	Accessibility was identified as a key advantage, whereas convenience was not consistently highlighted.	Personalised learning was highlighted as a key advantage.
H4: Ethical concerns such as privacy and bias will be key.	Ethical concerns were reported, but they did not significantly affect students' willingness to use GAI tools.	Results were similar to those in Europe.
H5: GAI tools will be most useful for resolving specific issues and supporting projects.	GAI tools were found to be most useful for written assignments and project support.	Similar results were observed as in Europe.

and guidance for specific subject-related questions, supporting research tasks within the educational program and improving experiential learning through example generation, content Customisation, and scenario creation. The findings highlight the potential of GAI tools to enhance both practical engagement and creative learning experiences, making it a valuable tool for academic support and educational advancement.

3 Comparative analysis of results obtained from students from Europe and Australia

This section presents a comparative analysis of the results obtained from students in Europe and Australia, focusing on similarities and differences in their attitudes, perceptions, and experiences with GAI tools in educational programs. The analysis aims to provide deeper insight into the factors influencing the acceptance and application of this technology in different parts of the world. The comparative results are summarised in Table 6.

Key findings from the comparative analysis:

1. Familiarity with Technology and Levels of Adoption:

- In both Europe and Australia, IT students demonstrated greater familiarity with GAI systems. However, the differences in familiarity between IT students and students from other fields were **not statistically significant**.
- Australian students were notably **less likely** to emphasise comfort as a significant factor in using GAI tools.

2. Perceived benefits:

- In Europe, **accessibility** was identified as the most significant advantage of GAI tools.
- In Australia, students valued **personalised learning** more prominently, suggesting a preference for tailored educational experiences.

3. Ethical concerns:

Although ethical concerns were reported in both regions, they were not found to significantly impact the willingness to adopt GAI tools. However, research highlights the interplay between algorithmic systems and commercial logics, such as targeted advertising, which can deepen disinformation risks (Díaz Ruiz, 2025). Privacy and security concerns were more prominent than bias or fairness issues, but they still had a limited influence on students' willingness to use the tool. Additionally, the survey findings revealed a notable gap in AI literacy programs, particularly in Australia, where institutional adoption remains ad hoc and inconsistent. Expanding formal AI literacy initiatives would be critical to ensure ethical and informed use of GAI tools across disciplines. Qualitative comments further illuminate regional distinctions. European students often framed GAI use within institutional or regulatory contexts, mentioning academic integrity policies or GDPR-related limitations, whereas Australian students tended to describe their use in terms of personal productivity and experimentation.

For example, one European student wrote, "Our professors warned us not to use ChatGPT in graded work, but I use it for brainstorming and summaries." In contrast, an Australian student observed, "ChatGPT helps me test ideas before I ask the lecturer. It's like a private tutor." These perspectives reflect differing educational cultures and regulatory climates that shape everyday practices.

- Privacy and security concerns were more prominent than bias or fairness issues, but they still had a limited influence on students' willingness to use the tool.

4. Key GAI applications:

- In both regions, GAI tools were identified as most useful for:

Supporting project development

Assisting with written assignments
Solving subject-specific problems

4 Discussion

This study set out to examine patterns in students' perceptions, ethical considerations, and readiness to adopt generative AI (GAI) tools in higher education across European and Australian contexts. Rather than interpreting the findings in terms of binary hypothesis confirmation, the discussion adopts an interpretative lens aligned with the study's exploratory and comparative design. The results reveal a consistent tension between perceived educational utility, ethical awareness, and actual adoption behaviour, offering insight into how students pragmatically negotiate the opportunities and risks associated with GAI use.

4.1 Perceived utility as a dominant driver of adoption

Across both regions, perceived educational utility emerges as the most stable and influential factor shaping students' engagement with GAI tools. Students consistently emphasise accessibility, efficiency, and support for academic tasks such as writing and project work, reinforcing prior survey-based findings that position GAI tools as instrumental academic resources (Wecks et al., 2024b). Importantly, these perceptions persist across disciplines and regions, suggesting that pragmatic learning benefits constitute a shared motivational foundation for adoption.

However, the findings also indicate that perceived utility is not uniformly distributed across all dimensions of use. While accessibility is consistently recognised as a key advantage, other anticipated benefits, such as personalised learning or enhanced convenience, receive weaker empirical support. This nuance challenges overly optimistic narratives surrounding GAI as a transformative learning technology and suggests that students primarily value these tools for task optimisation rather than deep pedagogical innovation. In this sense, adoption reflects pragmatic alignment with academic demands rather than an endorsement of GAI as a comprehensive learning partner.

4.2 Ethical awareness and the normalisation of risk

A central contribution of this study lies in its examination of the relationship between ethical concerns and adoption behaviour. The findings indicate that students do acknowledge ethical risks associated with GAI use, including data privacy, algorithmic bias, and overreliance. However, these concerns do not uniformly translate into reduced willingness to adopt GAI tools. Instead, the results may suggest patterns consistent with selective risk normalisation, whereby certain ethical risks appear to be perceived as acceptable trade-offs in exchange for perceived academic benefits. However, longitudinal evidence would be required to confirm the stability of such processes.

Notably, privacy-related concerns demonstrate a more consistent association with adoption behaviour than concerns related to bias or dependency. This pattern suggests that ethical sensitivity is not absent, but rather differentiated. Risks that are more tangible, institutionally regulated, or personally consequential appear to carry greater weight in students' decision-making than abstract or long-term epistemic risks. This finding aligns with qualitative responses in which students articulate awareness of ethical issues while simultaneously expressing confidence in their ability to manage or mitigate them through responsible use. The qualitative responses further suggest that students do not interpret ethical risks as abstract moral dilemmas but as situational trade-offs embedded in academic pressures. Ethical reasoning, therefore, appears to operate within performance-oriented environments rather than as an independent normative framework.

Rather than indicating ethical disengagement, this pattern reflects a pragmatic ethical stance shaped by institutional norms, assessment pressures, and the perceived inevitability of AI integration in higher education. Ethical reasoning thus appears to function contextually, embedded within everyday academic practices rather than guided solely by normative principles.

4.3 Epistemic authority and cognitive offloading

The limited behavioural impact of ethical concerns raises important questions about shifts in epistemic authority and learning practices. As GAI tools increasingly generate fluent and contextually appropriate outputs, students may recalibrate their trust in traditional academic sources and processes. The findings suggest that many students position GAI tools as legitimate auxiliary knowledge resources, particularly for clarification, drafting, and problem-solving tasks.

This dynamic is closely connected to concerns around cognitive offloading. While strategic use of GAI tools may support learning by reducing extraneous cognitive load, habitual reliance risks narrowing opportunities for critical engagement and independent reasoning. Importantly, the present findings indicate that students rarely frame their use of GAI in epistemic terms. Instead, adoption decisions are predominantly guided by functional considerations, with limited reflection on how GAI use might reshape knowledge construction or evaluative judgment.

These results underscore the need to interpret student engagement with GAI not as a simple deficit in ethical awareness, but as an adaptive response to evolving academic environments in which efficiency and performance are strongly incentivised. Epistemic risks, while recognised in principle, are often deprioritised in practice.

4.4 The role of AI literacy in mediating adoption practices

AI literacy emerges as a critical mechanism shaping how students balance perceived utility and ethical considerations. The comparable levels of AI familiarity across certain disciplines may reflect the diffusion of GAI tools beyond traditional technical domains.

As generative AI interfaces are increasingly intuitive and accessible, disciplinary barriers appear to diminish, suggesting that tool usability and platform design

may play a more significant role than formal curricular exposure. Students reporting higher familiarity with GAI tools also report greater readiness to adopt them, suggesting that exposure and confidence play an important role in normalising use. Students reporting higher familiarity with GAI tools also report greater readiness to adopt them, suggesting that exposure and confidence play an important role in normalising use. However, the findings also indicate that familiarity does not necessarily equate to critical engagement.

This distinction is significant. Without structured opportunities to develop critical and ethical dimensions of AI literacy, increased familiarity may reinforce instrumental use rather than reflective practice. The results therefore support calls for multidimensional AI literacy frameworks that extend beyond technical proficiency to include epistemic evaluation, ethical reasoning, and contextual awareness.

Rather than acting as a simple protective factor, AI literacy appears to mediate how students interpret risks and benefits. In the absence of explicit pedagogical guidance, literacy may facilitate more efficient use without necessarily fostering deeper ethical or epistemic reflection. This insight highlights the importance of embedding AI literacy within curriculum design and assessment practices rather than treating it as an optional or peripheral competence.

4.5 Regional context and educational cultures

Although the study does not treat Europe and Australia as homogeneous entities, the comparative analysis reveals meaningful contextual patterns. European students demonstrate greater sensitivity to privacy and data protection concerns, reflecting the influence of regulatory frameworks and institutional discourse around data governance. In contrast, Australian students' responses suggest a more pragmatic orientation, with ethical considerations exerting a weaker influence on reported adoption behaviour.

These differences should not be interpreted as fixed cultural traits, but rather as reflections of broader educational and regulatory environments that shape how ethical risks are framed and prioritised. The findings thus reinforce the importance of contextualising GAI adoption within specific institutional and policy settings, rather than assuming universal patterns of ethical engagement.

5 Implications

The findings of this study have several implications for theory, educational practice, and policy development in higher education. Importantly, these implications are grounded in observed empirical patterns rather than prescriptive assumptions, reflecting the exploratory and comparative nature of the research.

5.1 Theoretical implications

From a theoretical perspective, the study contributes to the growing body of research on generative AI adoption by demonstrating that perceived educational utility, ethi-

cal awareness, and AI literacy operate in a non-linear and context-dependent manner. While prior survey-based studies often assume that ethical concerns act as direct barriers to adoption, the present findings suggest a more nuanced relationship characterised by selective risk normalisation. This challenges models that position ethical concern as a uniformly inhibiting factor and highlights the need for theoretical frameworks that account for pragmatic trade-offs in student decision-making.

The integration of epistemic authority and cognitive offloading into the interpretation of student adoption behaviour extends existing technology adoption perspectives beyond instrumental use. The findings indicate that students' engagement with generative AI tools cannot be fully explained through traditional usefulness-oriented models alone. Instead, adoption reflects an adaptive negotiation shaped by performance pressures, institutional norms, and evolving conceptions of legitimate knowledge sources. This insight supports calls for adoption models that explicitly incorporate epistemic dimensions alongside ethical and literacy-related factors.

Furthermore, the study underscores the role of AI literacy as a mediating mechanism rather than a simple antecedent or outcome of AI use. Increased familiarity with generative AI tools appears to facilitate adoption, but does not necessarily promote critical or reflective engagement. This finding refines theoretical understandings of AI literacy by demonstrating that its impact depends on how literacy is conceptualised and embedded within educational contexts.

5.2 Practical implications for higher education institutions

The practical implications of the study are closely tied to the finding that students prioritise functional benefits such as accessibility and efficiency over abstract ethical considerations. Given that ethical concerns alone do not consistently deter adoption, institutions should not assume that awareness campaigns or policy statements are sufficient to guide responsible use. Instead, ethical and epistemic considerations need to be operationalised within teaching, assessment, and curriculum design.

Specifically, the finding that privacy-related concerns carry greater weight than other ethical risks suggests that students respond more strongly to risks that are clearly regulated and institutionally enforced. Higher education institutions may therefore achieve greater impact by aligning ethical guidance with concrete academic practices, such as data anonymisation protocols, clear rules regarding the use of AI tools in assessments, and transparent communication about data handling when interacting with large language models.

The results also highlight the need to move beyond generic AI literacy initiatives. While students reporting higher familiarity with GAI tools are more willing to adopt them, this familiarity does not automatically translate into a critical evaluation of AI-generated content. Institutions should therefore design AI literacy interventions that explicitly address epistemic risks, including verification strategies, bias recognition, and reflective use. Embedding such components into coursework, rather than offering standalone training sessions, may better align literacy development with students' actual learning practices. For example, institutions could develop short AI literacy modules integrated into first-year orientation courses, focusing on verification strategies, bias detection, and responsible prompting practices. In parallel, instructors

might implement structured AI-reflection assignments requiring students to document and critically evaluate how generative tools were used in their academic work.

5.3 Policy and context-sensitive implications

At the policy level, the cross-regional findings underscore the importance of context-sensitive approaches to generative AI governance in higher education. The observed differences between European and Australian students suggest that regulatory environments and institutional cultures shape how ethical risks are perceived and prioritised. Policies developed in highly regulated contexts may place greater emphasis on data protection and compliance, whereas more flexible environments may prioritise pedagogical innovation and practical integration.

These findings imply that one-size-fits-all policy frameworks are unlikely to be effective. Instead, policymakers should consider how national regulations, institutional norms, and disciplinary cultures interact to influence student behaviour. For example, policies that foreground data protection and academic integrity may resonate more strongly in contexts where such issues are already salient, while policies emphasising responsible innovation and guided experimentation may be more effective in environments oriented toward pragmatic adoption.

Finally, the study highlights the need for closer alignment between institutional policies and students' lived academic experiences. The persistence of adoption despite acknowledged ethical risks suggests that policies perceived as disconnected from everyday learning practices may have limited influence. Policymakers and institutional leaders should therefore prioritise mechanisms that integrate ethical considerations into routine academic workflows, ensuring that responsible use of generative AI is supported not only in principle, but also in practice.

6 Limitations and future research

Several limitations of this study should be acknowledged, as they directly shape the interpretation and scope of the findings. First, the study relies on convenience sampling, which constrains the generalisability of the results. Participants self-selected into the survey, and the sample is disproportionately composed of students from Information Technology and Computer Science disciplines. This disciplinary imbalance likely amplifies pragmatic perceptions of generative AI tools and may contribute to the observed normalisation of risks, particularly with respect to efficiency-oriented use. Consequently, the findings should not be interpreted as representative of the broader student population across all academic fields.

Second, the aggregation of data from multiple European countries into a single regional category introduces an additional limitation. While this approach enables cross-regional comparison with Australia, it inevitably obscures intra-regional diversity in educational cultures, institutional policies, and regulatory environments. Differences between Northern and Southern Europe, as well as between EU and non-EU contexts, may influence students' ethical sensitivity and adoption behaviour in ways that cannot be disentangled within the present design. As such, the regional findings

should be understood as indicative of broad contextual tendencies rather than precise cultural distinctions.

Third, the study is based on self-reported data, which introduces the possibility of social desirability bias and retrospective rationalisation. As with all self-reported survey data, the results may also be influenced by social desirability bias and self-enhancement effects. Students may underreport ethically questionable practices or overestimate their level of critical engagement with AI tools. Students' reported ethical awareness and readiness to adopt generative AI tools may not fully reflect their actual practices, particularly in assessment contexts where AI use is regulated or discouraged. This limitation is especially relevant given the finding that ethical concerns do not consistently translate into behavioural restraint, as reported attitudes may overestimate reflective engagement relative to real-world use.

Finally, the cross-sectional nature of the study limits its ability to capture temporal dynamics in students' perceptions and practices. Generative AI technologies, institutional policies, and pedagogical norms are evolving rapidly, and students' ethical reasoning and adoption behaviour are likely to change over time. The present findings therefore provide a snapshot rather than a developmental account of student engagement with generative AI in higher education.

These limitations point to several directions for future research. Longitudinal studies are needed to examine how students' perceptions, ethical concerns, and actual use of generative AI tools evolve as institutional policies mature and as students gain more sustained experience with these technologies. Given the rapidly evolving nature of generative AI technologies and institutional policies, longitudinal research designs are particularly important to examine whether ethical sensitivity increases, decreases, or stabilises over time as students accumulate experience with GAI tools. Future research would also benefit from stratified sampling strategies that ensure balanced representation across disciplines, allowing for more robust comparison of epistemic and ethical orientations beyond technically oriented fields. Additionally, disaggregated regional analyses within Europe could provide deeper insight into how specific regulatory and cultural contexts shape student engagement with generative AI. Finally, mixed-method designs that integrate behavioural data, classroom observations, or learning analytics with self-reported measures could offer a more comprehensive understanding of how ethical awareness and AI literacy translate into practice.

7 Conclusion

This study examined students' perceptions, ethical considerations, and readiness to adopt generative AI (GAI) tools in higher education, drawing on a comparative analysis of European and Australian contexts. Rather than approaching adoption as a purely technical or ethical decision, the findings highlight how students pragmatically negotiate perceived educational benefits and associated risks within their academic environments.

Across both regions, perceived educational utility emerges as a consistent driver of engagement with GAI tools, particularly in relation to accessibility, efficiency, and

support for academic tasks. While students demonstrate awareness of ethical concerns such as privacy, bias, and overreliance, these concerns do not uniformly translate into reduced willingness to adopt GAI tools. Instead, the results may indicate patterns consistent with selective risk normalisation, whereby certain risks appear to be deprioritised in favour of perceived academic benefits. Given the cross-sectional design, these interpretations should be understood as indicative rather than conclusive.

Importantly, the study contributes to existing literature by moving beyond binary assumptions about ethical concern as a barrier to adoption. The findings indicate that ethical awareness, AI literacy, and adoption behaviour interact in context-dependent ways, shaped by institutional norms, regulatory environments, and students' familiarity with GAI systems. Differences observed between European and Australian students further underscore the importance of situating student engagement with generative AI within specific educational and policy contexts, rather than assuming homogeneous patterns of use.

The contribution of this study lies in its empirically grounded insight into how students balance efficiency, responsibility, and epistemic judgment when engaging with generative AI tools. By framing adoption as a pragmatic negotiation rather than a linear ethical decision, the study offers a more nuanced understanding of student behaviour in AI-mediated learning environments.

These conclusions should be interpreted in light of the study's methodological constraints, including its cross-sectional design and sampling strategy. Nevertheless, the findings provide a timely foundation for future research that seeks to examine how students' perceptions and practices evolve as generative AI technologies, institutional policies, and educational norms continue to develop. Longitudinal and context-sensitive studies will be essential for advancing understanding of responsible and sustainable integration of generative AI in higher education.

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Declaration

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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